Explainable Reinforcement Learning for Emergency-Aware Traffic Signal Optimisation

Puranjay Singh[1], Chirag Verma[2], Akshat Abhivardan[3]
[1,2,3]Department of Computer Science and Engineering, Bennett University, Greater Noida, India
[1]puranjaysingh222@gmail.com [2]vchirag012@gmail.com [3]akshat@gmail.com

***Abstract*—Emergency vehicle delay at signalised intersections remains difficult to manage under dense urban traffic conditions. Conventional traffic controllers and several reinforcement learning approaches primarily optimise average traffic flow and do not explicitly prioritise emergency vehicles during congestion-heavy scenarios.**

This work proposes APA-DQN-V2, an Adaptive Priority Attention Deep Q-Network for emergency-aware traffic signal optimisation. The framework combines adaptive feature weighting with a dueling Q-network structure to balance emergency delay, normal waiting time, and queue growth. Experiments were conducted in the SUMO traffic simulator under multiple traffic conditions, including emergency-heavy and unseen scenarios.

Initial experiments using standard DQN produced unstable phase transitions during high-density traffic conditions. The proposed attention mechanism improved emergency response consistency while maintaining lower queue growth compared to baseline approaches.

Integrated Gradients and counterfactual analysis were further used to examine feature sensitivity and policy behaviour. Emergency-related features consistently produced higher attribution scores during emergency scenarios, indicating that the learned policy relied strongly on emergency-aware inputs.

***Index Terms*—Reinforcement Learning, Traffic Signal Control, Emergency Vehicles, Explainable AI, Deep Q-Network, Intelligent Transportation Systems**

I. Introduction

Traffic congestion in cities is a serious issue as a result of increased urbanization. This becomes an even more serious problem especially when dealing with emergency situations since delay at a controlled intersection can prove fatal. It therefore makes sense for any traffic signal system to give priority to emergency vehicles.

However, traditional methods of traffic control depend on predetermined timings for the signals which do not allow for effective adaptation to different degrees of congestion on the roads. Even though actuated systems adapt to changing conditions to some extent by receiving input from sensors, they still depend on predetermined algorithms.

Reinforcement learning techniques have recently been explored for adaptive traffic control in adaptive traffic management because they can continuously improve decision-making through interactions with changing environments [1], [7], [8]. RL-based approaches enable real-time adaptation of signal phases based on observed traffic conditions. However, most existing methods focus on single-objective optimization, such as minimizing waiting time or queue length [10], [11], and do not explicitly address emergency vehicle prioritization. Furthermore, many RL models operate as black-box systems, limiting interpretability in safety-critical applications [5], [14].

In order to solve the above shortcomings, this paper introduces the APA-DQN-V2 framework, which incorporates the priority-aware attention model into the dueling deep Q-learning network architecture. This framework enables dynamic re-weighting of traffic parameters according to the emergency context, allowing efficient prioritization without affecting the traffic efficiency. Unlike other methods, the APA-DQN-V2 framework combines the priority-aware attention model with explainable reinforcement learning, thereby offering enhanced performance as well as interpretability of the emergency-aware traffic control system.

The valuable insights of the work are summarised below:

- Adaptive Priority Attention mechanism for dynamic feature weighting in traffic signal control
- A MORL(multi-objective reinforcement learning) framework balancing emergency delay, normal waiting time, and congestion
- Explainable AI techniques to improve the transparency and interpretability of decisions
- Evaluation of models on five different scenarios demonstrating generalisation for random or unexpected traffic conditions

II. Literature Review

The use of reinforcement learning in adaptive traffic signal control has gained popularity owing to its capacity to determine optimal policies in changing environments. Early research by Mnih et al. [1] established the potential of Deep Q-Networks (DQN) in tackling complicated decision-making problems, which subsequently paved the way for its application in traffic systems.

Many studies have used RL techniques in traffic signal optimisation [6]–[8]. Liang et al. [7] and Wang et al. [8] suggested RL models for traffic signal optimisation that enhance traffic flow through varying signal phases. Nonetheless, these methods mainly concentrate on decreasing average waiting time or queue length without considering the requirement for emergency vehicle prioritisation.

The dueling architecture proposed by Wang et al. [2], among other advanced deep architectures, has contributed to stable learning processes through the decoupling of value and advantage functions. Policy gradient methods like PPO [3]

TABLE I
KEY DIFFERENCES BETWEEN EXISTING WORK AND APA-DQN-V2

Ref.	Limitation	Improvement
[1], [2]	Generic RL, no traffic prioritization	Attention-based emergency-aware DQN
[3]	High complexity, weak emergency handling	Balanced delay and congestion control
[6]–[8]	No explicit emergency handling	Priority-aware feature weighting
[9], [16]	Complex multi-agent coordination	Simpler single-agent prioritization
[11]–[13]	Difficult multi-objective trade-offs	Weighted reward + adaptive attention
[14], [17]	Lack of interpretability	Integrated Gradients + counterfactuals

provide better convergence in continuous environments. Multi-agent RL algorithms [9], [16] have been developed to facilitate traffic management in extensive traffic networks.

The emergence of multi-objective optimisation paradigms in recent literature [11]–[13] offers a solution to conflicting traffic objective functions. However, a proper trade-off between the prioritisation of emergencies and other traffic parameters is still lacking. In addition, RL-based traffic control approaches have been treated mostly as black-box methods, thus limiting their use in mission-critical applications.

Various Explainable AI (XAI) models have been employed extensively in machine learning to address interpretability issues [5]. Nevertheless, their application in RL-based traffic signal controllers is still underexplored.

Unlike previous research, the proposed framework incorporates a novel attention-based prioritisation model, which considers feature importance during emergencies alongside multi-objective optimisation and XAI within an RL framework.

Table I provides a structured comparison of existing approaches and highlights the improvements introduced by APA-DQN-V2.

III. PRELIMINARIES

A. Markov Decision Process

Traffic signal control is formulated as a Markov Decision Process (MDP) [4], where the agent observes the current traffic state and selects signal phase transitions to optimise long-term traffic flow. The state space S contains traffic-related observations, while A denotes the set of valid signal actions. The transition dynamics are represented by P, and R defines the reward obtained after each action.

The discounted return is defined as:

$$Q^\pi(s,a) = \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t r_t\right] \tag{1}$$

A discounted formulation was selected to avoid aggressive short-term phase switching that occasionally increased congestion during early simulation experiments.

B. Deep Q-Network

A Deep Q-Network (DQN) [1] is used to estimate state-action values directly from traffic observations:

$$Q(s,a;\theta) \tag{2}$$

The network parameters are updated using temporal-difference learning:

$$L(\theta) = \mathbb{E}\left[\left(r + \gamma \max_{a'} Q(s',a';\theta^-) - Q(s,a;\theta)\right)^2\right] \tag{3}$$

Initial experiments with direct Q-value updates produced unstable learning behaviour under dense traffic conditions. A target network was therefore maintained to stabilise training.

C. Dueling Architecture

The dueling architecture proposed by Wang et al. [2] separates state evaluation from action-specific advantage estimation:

$$Q(s,a) = V(s) + A(s,a) - \frac{1}{|\mathcal{A}|}\sum_{a'} A(s,a') \tag{4}$$

In preliminary simulations, multiple signal transitions often produced similar short-term rewards. The dueling structure improved action differentiation during emergency-heavy traffic scenarios.

D. Multi-Objective Optimization

Emergency-aware traffic control introduces competing objectives, since aggressive prioritisation of emergency vehicles can negatively affect normal traffic flow. The reward formulation therefore combines emergency delay, regular waiting time, and queue length into a weighted optimisation objective.

E. Explainable Reinforcement Learning

Interpretability becomes important in emergency traffic systems because policy decisions directly influence vehicle movement under time-sensitive conditions. Integrated Gradients [18] and counterfactual analysis were used to examine feature sensitivity and analyse how emergency-related inputs influenced action selection.

IV. PROPOSED METHODOLOGY

The proposed APA-DQN-V2 framework was developed to reduce emergency vehicle delay without heavily increasing congestion for regular traffic. Early experiments using standard DQN [1] produced unstable phase switching under dense traffic conditions, particularly when emergency vehicles appeared frequently. To address this, an adaptive attention mechanism and dueling Q-network structure [2] were incorporated into the control pipeline.

The overall architecture is shown in Fig. 1. Traffic-state features are encoded first and then passed through the attention layer, where emergency-related inputs receive higher importance during critical traffic conditions. The weighted feature

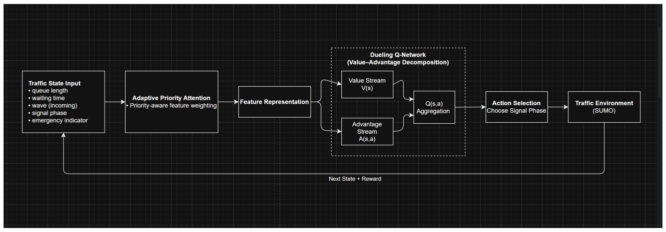

Fig. 1. Architecture of the proposed APA-DQN-V2 framework.

representation is then passed to the dueling network, where separate streams estimate state value and action advantage.

During simulation, the attention layer reduced unnecessary phase changes in low-priority lanes while improving response time for emergency vehicles. Q-values were updated continuously using feedback obtained after each environment interaction.

A. State Representation

The state vector includes queue length, waiting time, current signal phase, and emergency vehicle information. Emergency presence and distance were included because initial simulations showed that queue-based features alone were insufficient during emergency-heavy traffic conditions.

B. Action Space

At each decision step, the agent selects the next traffic signal phase from a predefined set of valid phase transitions. Direct phase control was preferred over fixed timing because traffic density varied significantly across scenarios.

C. Adaptive Priority Attention Mechanism

Emergency vehicle features were weighted dynamically using an adaptive attention mechanism. During early experiments, the agent frequently prioritised congested lanes even when emergency vehicles were present nearby. Increasing the contribution of emergency-related inputs improved response consistency during those cases.

$$\tilde{s}_i = w_i \cdot s_i \tag{5}$$

where w_i denotes the learned importance assigned to feature s_i. Emergency-presence and distance features received larger weights during emergency scenarios.

D. Dueling Network Architecture

The dueling structure was adopted because standard DQN frequently produced similar Q-values for multiple signal transitions under heavy congestion. Separating state-value estimation from action advantage improved action differentiation during these scenarios.

E. Reward Function

The reward function combines emergency delay, normal waiting time, and queue length:

$$R = -\alpha WT_{emergency} - \beta WT_{normal} - \gamma Queue \tag{6}$$

Large emergency weights reduced ambulance delay but occasionally increased congestion in neighbouring lanes during preliminary simulations. The final weighting configuration therefore prioritised emergency traffic while maintaining stable queue growth.

F. Training Procedure

Training was performed using experience replay and a target network [1]. An ϵ-greedy exploration strategy was used during early episodes to increase state-space coverage before gradually shifting toward exploitation.

Replay-based updates reduced instability under highly variable traffic arrivals.

Algorithm 1: APA-DQN-V2 Training Procedure

```
Initialise Q-network with parameters θ
Initialise target network θ⁻
for each episode do
   Observe initial state s
   for each step do
      Select action a using ε-greedy policy
      Execute action and observe reward r and next state s′
      Store transition (s, a, r, s′) in replay buffer
      Sample mini-batch and update θ
      Periodically update target network θ⁻
      s ← s′
   end for
end for
```

V. Experimental Setup

A. Simulation Environment

Experiments are conducted using Simulation of Urban Mobility (SUMO), a widely used microscopic traffic simulation platform [19]. SUMO provides realistic modelling of vehicle dynamics, traffic flow, and signal control, making it suitable for evaluating traffic optimisation algorithms.

B. State Representation

The state representation is consistent with the formulation described in the methodology and includes:

- Queue length at each lane
- Waiting time of vehicles
- Current signal phase
- Emergency vehicle presence and distance

C. Evaluation Scenarios

The model was evaluated under multiple traffic settings to examine its behaviour under both normal and emergency-heavy conditions:

- **Baseline:** Regular traffic flow with moderate vehicle density
- **Low Demand:** Sparse traffic with fewer vehicle arrivals
- **High Demand:** Congested traffic with longer queue formation
- **Emergency Stress:** Frequent emergency vehicle arrivals within short intervals
- **Unseen Scenario:** Traffic distributions not encountered during training

High-density and emergency-stress scenarios produced the largest variation in queue growth during preliminary testing. The unseen scenario was included to examine whether the learned policy remained stable under traffic patterns outside the training distribution.

D. Baseline Models

The proposed model is compared against the following baselines:

- Fixed-Time Control
- Deep Q-Network (DQN) [1]
- Dueling-DQN [2]
- Proximal Policy Optimisation (PPO) [3]
- PAE-DQN

These baselines represent both traditional and reinforcement learning-based approaches for traffic signal control.

E. Evaluation Metrics

Performance is evaluated using standard traffic control metrics commonly adopted in reinforcement learning-based traffic signal optimisation [7], [8]. The following metrics are used:

- Emergency vehicle delay
- Average waiting time of normal vehicles
- Average queue length

These metrics collectively capture the trade-off between emergency prioritisation, overall traffic efficiency, and congestion control, providing a comprehensive evaluation of model performance.

F. Training Configuration

The model is trained for 500 episodes with a learning rate of 0.001 and discount factor $\gamma = 0.99$. The replay buffer size is set to 10,000, and the batch size is 64. The exploration rate ϵ decays from 1.0 to 0.05 over the training process to ensure a balance between exploration and exploitation.

VI. Results and Discussion

A. Model Comparison

APA-DQN-V2 model lowers the average waiting time for normal vehicles by about 65%, relative to the DQN model, and lowers the average queue length by 70%, relative to the PPO model, indicating better performance.

TABLE II
PERFORMANCE COMPARISON ACROSS MODELS. APA-DQN-V2 ACHIEVES THE BEST OVERALL TRADE-OFF BETWEEN EMERGENCY DELAY, NORMAL WAITING TIME, AND QUEUE LENGTH.

Model	Emergency Delay ↓	Normal Wait ↓	Queue Length ↓
DQN	275	8.5	12.0
Dueling-DQN	77	11.0	9.8
PPO	430	18.5	14.0
PAE-DQN	480	3.5	4.2
APA-DQN-V2	120	**3.0**	**3.4**

Though the Dueling-DQN algorithm has the best emergency delay, it causes more waiting time and congestion. On the other hand, APA-DQN-V2 ensures low waiting time and queue length, in addition to effective emergency prioritisation.

B. Performance Comparison

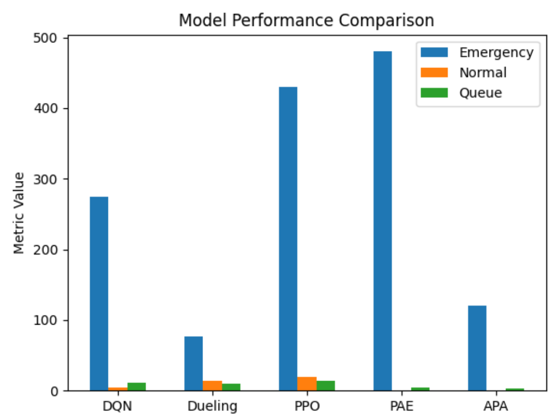

Fig. 2. Comparison of models across key traffic metrics.

As shown in Fig. 2, APA-DQN-V2 consistently performs well across all evaluation metrics, avoiding extreme trade-offs observed in baseline models.

C. Scenario Generalization

The results based on scenarios indicate that the APA-DQN-V2 algorithm adjusts efficiently according to changes in traffic. The emergency delay rises in stressful scenarios due to high demand levels; however, it stays manageable compared to existing methods. The normal waiting time stays constant, proving that prioritising emergency vehicles does not affect regular traffic significantly. The queue length differs depending on the traffic density level, but it stays reasonable.

In addition, the algorithm provides good performance in unseen scenarios, showing its excellent generalization capabilities in unpredictable traffic conditions.

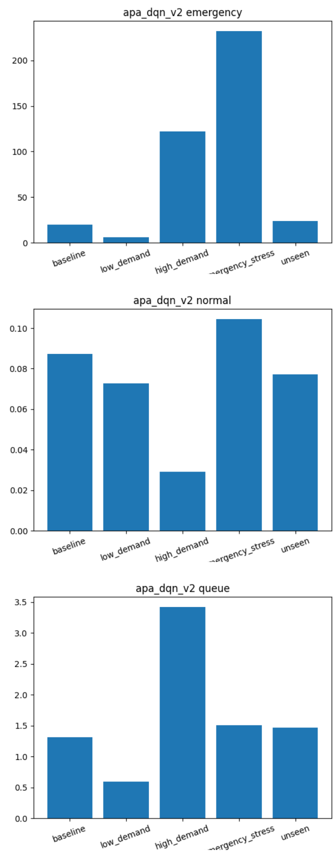

Fig. 3. Scenario-wise performance of APA-DQN-V2 across emergency delay, normal waiting time, and queue length.

D. Trade-off Analysis

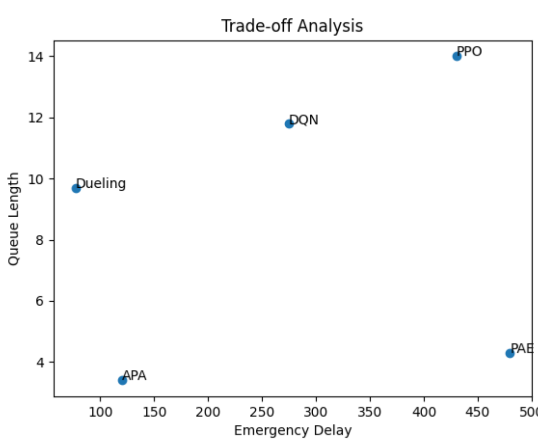

Fig. 4. Trade-off between emergency delay and queue length.

APA-DQN-V2 operates close to the Pareto-optimal region, demonstrating an effective balance between emergency prioritization and congestion control.

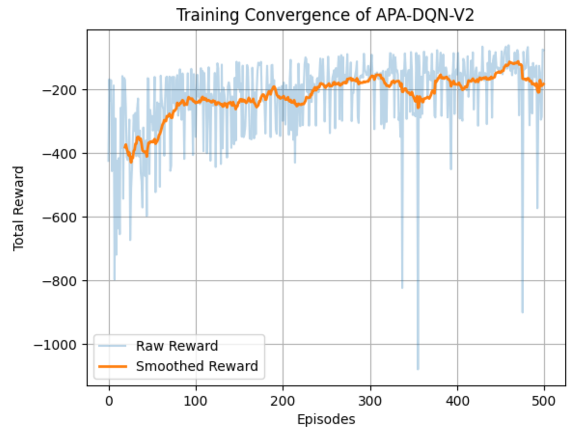

Fig. 5. Training convergence of APA-DQN-V2. The smoothed reward curve shows stable learning with reduced variance across episodes.

E. Training Behavior

As shown in Fig. 5, the smoothed reward curve demonstrates stable convergence with reduced variance, indicating effective policy learning and robustness under stochastic traffic conditions.

Compared to baseline models, the proposed approach exhibits:

- Faster convergence due to structured value–advantage learning
- Reduced oscillations in reward progression
- Improved stability under varying traffic conditions

These observations highlight the effectiveness of the adaptive priority attention mechanism in guiding the learning process. The reduced variance in the smoothed reward curve further indicates improved policy stability and robustness under stochastic traffic conditions.

VII. Explainable Reinforcement Learning Analysis

Policy behaviour during emergency prioritisation was difficult to interpret directly from Q-values alone. To analyse feature sensitivity and action preference, attribution-based and counterfactual explainability methods were applied to APA-DQN-V2.

A. Feature Importance

Integrated Gradients (IG) [18] was used to estimate the contribution of individual state variables to Q-value estimation:

$$IG_i = (s_i - s'_i) \int_0^1 \frac{\partial Q(s' + \alpha(s - s'))}{\partial s_i} d\alpha \tag{7}$$

Emergency-presence and distance features consistently produced larger attribution scores than standard traffic-density features during emergency-heavy simulations. Queue-length features remained dominant during non-emergency traffic conditions.

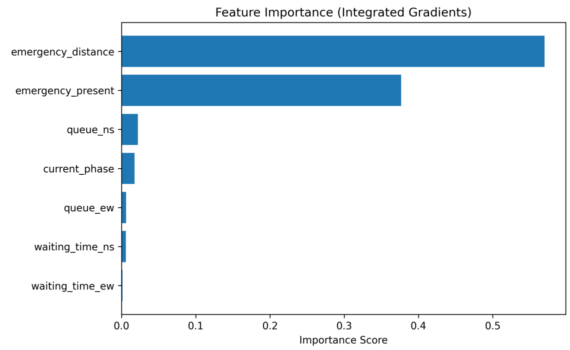

Fig. 6. Feature attribution scores obtained using Integrated Gradients under emergency-aware traffic conditions.

B. Value–Advantage Decomposition

The dueling architecture [2] separates state-value estimation from action-specific advantage estimation. During emergency scenarios, the advantage stream produced higher variance between competing actions compared to normal traffic conditions.

Fig. 7 shows that emergency scenarios resulted in stronger separation between candidate signal actions, indicating more decisive policy behaviour under time-sensitive traffic conditions.

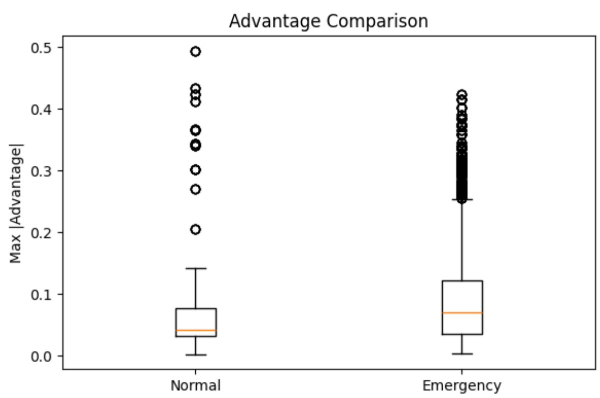

Fig. 7. Comparison of advantage values under normal and emergency traffic conditions.

C. Counterfactual Analysis

Counterfactual perturbations were applied to emergency-related inputs, including emergency distance and vehicle presence indicators. Small perturbations in these features frequently altered phase-selection behaviour, particularly during congested intersections with competing traffic demand.

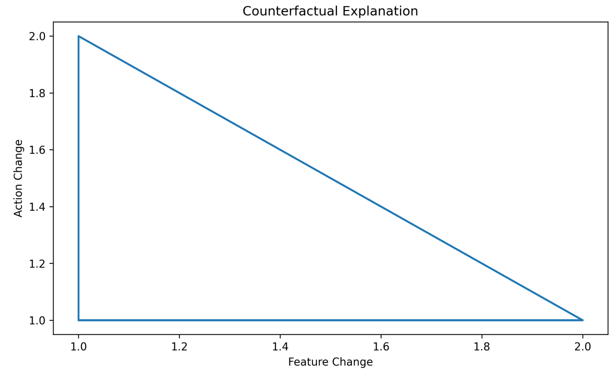

Fig. 8. Counterfactual analysis illustrating policy sensitivity to emergency-related inputs.

D. Interpretability Insights

The explainability analysis indicated that emergency-related features strongly influenced action selection during critical traffic conditions. The dueling structure also produced clearer separation between competing actions during emergency arrivals compared to normal traffic flow.

Although the policy remained relatively stable across most evaluation scenarios, larger variance in action preference was observed during emergency-heavy simulations.

VIII. Limitations

The proposed method will need to undergo a real-life traffic system test in addition to its assessment in a SUMO-based simulator environment. In addition, the time required by the training phase in the process will be considerable due to the interaction with the simulator environment. Scaling up the methodology to multiple intersections can also serve as a basis for future work.

IX. Conclusion

This paper presents APA-DQN-V2, an explainable reinforcement learning framework for emergency-aware traffic signal optimisation. The proposed model integrates an adaptive priority attention mechanism with a dueling architecture to effectively balance emergency vehicle prioritisation and overall traffic efficiency.

Experiment results show that the APA-DQN-V2 model provides better performance among the above three criteria compared to baseline models. In addition, it shows better adaptability to various traffic situations, which indicates good generalization capability of this model under varying traffic conditions.

Also, incorporation of explainable AI techniques helps bring clarity into decision-making, which is a weakness that was identified with reinforcement learning algorithms.

On the whole, the developed strategy has proven to be quite effective in minimizing the delay of an emergency vehicle while ensuring stable traffic flow.

Future work will focus on extending the model to multi-intersection traffic networks and integrating real-time sensor data for deployment in real-world smart city environments.

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